Another variety of radio telescope is the interferometer, comprising arrays of identical dish antennas, spread across swaths of countryside and electronically linked to work in concert. The result is a single, coherent, super-high-resolution image of radio-emitting cosmic objects. Although "supersize me" was the unwritten motto for telescopes long before the fast food industry coined the slogan, radio interferometers form a jumbo class unto themselves. One of them, a very large array of radio dishes near Socorro, New Mexico, is officially called the Very Large Array, with twenty-seven eighty-two-foot dishes positioned on tracks crossing twenty-two miles of desert plains. This observatory is so cosmogenic, it has appeared as a backdrop in the films *2010: The Year We Make Contact* (1984), *Contact* (1997), and *Transformers* (2007). There's also the Very Long Baseline Array, with ten eighty-two-foot dishes spanning 5,000 miles from Hawaii to the Virgin Islands, enabling the highest resolution of any radio telescope in the world.

In the microwave band, relatively new to interferometers, we've got the sixty-six antennas of ALMA, the Atacama Large Millimeter Array, in the remote Andes Mountains of northern Chile. Tuned for wavelengths that range from fractions of a millimeter to several centimeters, ALMA gives astrophysicists high-resolution access to categories of cosmic action unseen in other bands, such as the structure of collapsing gas clouds as they become nurseries from which stars are born. ALMA's location is, by intention, the most arid landscape on Earth—three miles above sea level and well above the wettest clouds. Water may be fine for microwave cooking but it's bad for astrophysicists, because the water vapor in Earth's atmosphere chews up pristine microwave signals from across the galaxy and beyond. These two phenomena are, of course, related: water is the most common ingredient in food, and microwave ovens primarily heat water. Taken together, you get the best indication that water absorbs microwave frequencies. So if you want clean observations of cosmic objects, you must minimize the amount of water vapor between your telescope and the universe, just as ALMA has done.

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At the ultrashort-wavelength end of the electromagnetic spectrum you find the high-frequency, high-energy gamma rays, with wavelengths measured in picometers.[†][†][†][†][†][†] Discovered in 1900, they were not detected from space until a new kind of telescope was placed aboard NASA's *Explorer XI* satellite in 1961.